

Recurso a Fondo

Tema 9: Resolución de ecuaciones diferenciales estocásticas y cálculo de momentos

ECUACIONES DIFERENCIALES ESTOCÁSTICAS

Practice more about some SDEs

Apply the technique exhibited in the previous exercise to solve the following SDE and to compute the mean and the variance of the solution. Check the results with Mathematica.

$$\begin{cases} dX(t) &= dt + 2\sqrt{X(t)}dW(t) \\ X(0) &= x_0 \end{cases} \quad (1)$$

To solve the problem introduced in Equation 1 we must recall Itô's Lemma (III) in differential form. Itô's Lemma states that a given continuous function $f(t, x) \in \mathcal{C}^{1,2}$ is a solution of a problem of the form,

$$dY(t) = A^{(1)}(t, Y(t))dt + A^{(2)}(t, Y(t))dW(t) \quad (2)$$

if it satisfies the following expression:

$$df(t, x) = \left[\frac{\partial f(t, x)}{\partial t} + A^{(1)}(t, Y(t)) \frac{\partial f(t, x)}{\partial x} + \frac{1}{2} (A^{(2)}(t, Y(t)))^2 \frac{\partial^2 f(t, x)}{\partial x^2} \right] dt + A^{(2)}(t, Y(t)) \frac{\partial f(t, x)}{\partial x} dW(t) \quad (3)$$

where, $A^{(1)}(t, Y(t)), A^{(2)}(t, Y(t))$ are the coefficients characterising the SDE.

Applying Equation 3 to the problem introduced in the exercise (Equation 1), we get:

$$df(t, x) = \left[\frac{\partial f(t, x)}{\partial t} + \frac{\partial f(t, x)}{\partial x} + 2x \frac{\partial^2 f(t, x)}{\partial x^2} \right] dt + 2\sqrt{x} \frac{\partial f(t, x)}{\partial x} dW(t) \quad (4)$$

At this stage, we need to select a function $f(t, x)$. The idea is to obtain a expression where the differential terms only depend on t . Thus, we will select $f(t, x)$ to eliminate x -terms.

We choose,

$$f(t, x) = p(t)\sqrt{x} + q(t) \quad (5)$$

where $p(t), q(t)$ are time-dependent functions to be particularised later on. The derivative terms introduced in Equation 4 are given by:

$$\begin{aligned} \frac{\partial f(t, x)}{\partial t} &= \frac{dp(t)}{dt} \sqrt{x} + \frac{dq(t)}{dt} \\ \frac{\partial f(t, x)}{\partial x} &= \frac{p(t)}{2\sqrt{x}}, \quad \frac{\partial^2 f(t, x)}{\partial x^2} = -\frac{p(t)}{4x^{\frac{3}{2}}} \end{aligned} \quad (6)$$

Replacing the derivatives terms (Equation 6) in Equation 4, we get:

$$\begin{aligned} d(p(t)\sqrt{x} + q(t)) &= \left[\frac{dp(t)}{dt}\sqrt{x} + \frac{dq(t)}{dt} + \frac{p(t)}{2\sqrt{x}} - \frac{p(t)}{2\sqrt{x}} \right] dt + p(t)dW(t) \\ &= \left[\frac{dp(t)}{dt}\sqrt{x} + \frac{dq(t)}{dt} \right] dt + p(t)dW(t) \end{aligned} \quad (7)$$

Since Equation 7 is verified for any functions $p(t), q(t)$, it will be verified for:

$$\begin{cases} p(t) &= 1 \\ g(t) &= 0 \end{cases} \quad (8)$$

Thus, Equation 7 is simplified as,

$$d(\sqrt{x}) = dW(t) \quad (9)$$

Taking integrals in Equation 9 we find:

$$\begin{aligned} \sqrt{X(t)} &= W(t) + C_1 \\ \implies X(t) &= (W(t) + C_1)^2 \end{aligned} \quad (10)$$

Applying the initial condition one can get:

$$X(0) = x_0 \Leftrightarrow \underbrace{(W(0) + C_1)^2}_{=0} = x_0 \implies C_1 = \sqrt{x_0} \quad (11)$$

Finally the solution of the SDE proposed in Equation 1 is given by:

$$\boxed{X(t) = (W(t) + \sqrt{x_0})^2} \quad (12)$$

Having solved the SDE we can compute the expectation and variance. Applying the expectation operator,

$$\mathbb{E}[X(t)] = \mathbb{E}[(W(t) + \sqrt{x_0})^2] \quad (13)$$

Recall the fundamental properties of the expectation operator:

$$\begin{aligned} \mathbb{E}[X(t)] &= \mathbb{E}[W^2(t)] + \mathbb{E}[2W(t)\sqrt{x_0}] + \mathbb{E}[x_0] \\ &= \mathbb{V}[W(t)] + 2\mathbb{E}[W(t)\sqrt{x_0}] + \mathbb{E}[x_0] \end{aligned} \quad (14)$$

Since x_0 is deterministic its expectation will be itself. Thus,

$$\mathbb{E}[X(t)] = \underbrace{\mathbb{V}[W(t)]}_{=t} + 2\sqrt{x_0} \underbrace{\mathbb{E}[W(t)]}_{=0} + x_0 \quad (15)$$

Finally, the expectation is given by:

$$\boxed{\mathbb{E}[X(t)] = t + x_0} \quad (16)$$

As far as the variance is concerned, we recall the relation between the variance and expectation and apply it to the stochastic process here analysed:

$$\begin{aligned} \mathbb{V}[X(t)] &= \mathbb{E}[X^2(t)] - (\mathbb{E}[X(t)])^2 \\ &= \mathbb{E}[(W(t) + \sqrt{x_0})^4] - (t + x_0)^2 \\ &= \mathbb{E}[W^4(t)] + 4\sqrt{x_0}\mathbb{E}[W^3(t)] + 6x_0\mathbb{E}[W^2(t)] + 4x_0^{3/2}\mathbb{E}[W(t)] + \mathbb{E}[x_0^2] - (t + x_0)^2 \end{aligned} \quad (17)$$

In Equation 17 we see that the variance of the stochastic process here evaluated depends only on deterministic variables and the moments of the standard Wiener process. We know that the moments of the standard Wiener process are given by:

$$\begin{aligned}\mathbb{E}[W^2(t)] &= \mathbb{V}[X(t)] = t \\ \mathbb{E}[W^3(t)] &= 0 \\ \mathbb{E}[W^4(t)] &= 3t^2\end{aligned}\tag{18}$$

Replacing the aforementioned terms in Equation 17 we find

$$\boxed{\mathbb{V}[X(t)] = 2(t^2 + 2x_0t)}\tag{19}$$

In conclusion the solution of Equation 1 is characterised by:

$$\boxed{\begin{cases} X(t) &= (W(t) + \sqrt{x_0})^2 \\ \mathbb{E}[X(t)] &= t + x_0 \\ \mathbb{V}[X(t)] &= 2(t^2 + 2x_0t) \end{cases}}\tag{20}$$